Vipin Kaushik

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Education

Indian Institute of Technology Kharagpur *Kharagpur, India*
B.Tech and M.Tech in Chemical Engineering (Dual Degree) *November 2022 to April 2027*

Experience

Full-Stack Developer Intern - Clairvyn AI **May 2026 to Jul 2026**
AI-powered floor planning platform for intelligent space planning, automated design, and streamlined architectural decision-making
- Engineered **15+** responsive **Next.js** UI panels, optimizing data routing for an AI-powered floor planning platform at scale.
- Integrated **Paytm PG APIs** to construct a secure subscription paywall, enabling seamless transactions for **150+** beta users.
- Automated **AWS EC2** deployments using **CI/CD** pipelines by GitHub Actions & **Docker**, cutting deployment time by **40%**.
- Refactored Node.js data pipelines, eliminating **N+1** queries and batching requests and cutting down the API latency by **50%**.

Frontend Engineering Intern - Unnati **Mar 2025 to May 2025**
SME platform for local Indian business owners, streamlining digital onboarding, connection and customer engagement
- Developed an **admin dashboard** with reusable **React** components and **hooks** for multi-step registration serving **30+** users.
- Built a reusable **design system**, standardizing **UI** components for consistent frontend development across core workflows.
- Defined frontend component architecture from insights across **20+** user interviews, creating scalable implementation plans.
- Engineered accessible **React** components meeting **WCAG 2.1**, ensuring **100%** screen-reader compatibility across interfaces.

Projects

AppForge - Next.js 15, Groq, Supabase, Clerk, Redis, Kafka, Zod **May 2026 to Aug 2026**
Metadata-driven deterministic application runtime *Link:* github.com/vpn-cli/appforge
- Accelerated dynamic UI generation to approx.**1.5s** by building a **Next.js 15** runtime, compiling LLM-streamed **JSON AST**s.
- Reduced prompt-to-layout latency to **sub 800ms** by engineering a deterministic Groq and Gemini dual-provider LLM pipeline.
- Mitigated **95%+** of AI hallucination crashes by utilizing the **Zod** schema validation within a fault-tolerant React registry.
- Increased **AI request throughput** via decoupling asynchronous tasks via a **Kafka queue** for scalable request processing.

Sportz - Next.js 15, React, Node.js, WebSockets, PostgreSQL, Drizzle **Mar 2025 to Apr 2025**
Real-time analytical platform to process scalable sports data *Link:* github.com/vpn-cli/sockets
- Engineered a real-time React/Node sports dashboard serving **under 80ms** updates, cutting latency **65%** via **WebSocket**.
- Improved PostgreSQL query execution by **40%** with **100%** type safety using **Drizzle ORM** schemas and **Zod** validation.
- Secured production Express.js APIs against bot traffic, maintaining **99.9%** uptime via **Arcjet** middleware & strict **CORS**.
- Cut the deployment time by **60%** with zero downtime by automating **CI/CD** for Vite on Vercel and **Node.js** on **Render**.

Technical Skills

Languages: JavaScript, TypeScript, C++, SQL
Backend: Node.js, Express.js, PostgreSQL, Supabase, Redis
Frontend: React.js, Next.js, TailwindCSS, GSAP
Tools: Git, Figma, Postman, Docker, AWS EC2, Paytm PG
Concepts: REST APIs, OOPs, System Design, Message Queues, Authentication, Real-time Systems, Microservices, CI/CD

Leadership & Activities

Design Coordinator - Illumination Magazine, IIT Kharagpur **Oct 2024 to Dec 2024**
- Directed visual design for **25+** annual magazine assets, strengthening the visual communication for **15,000+** readers.
- Guided **designers & editors** by establishing **clear** workflows, streamlining collaboration & ensuring **consistent** publication.

Design Team Head - Manga & Anime Society, IIT Kharagpur **Aug 2024 to Apr 2025**
- Led a creative team to produce **20+** posters, banners, and UI mockups for university-wide events, ensuring consistent branding.
- Conducted Photoshop, Figma workshops for junior designers, strengthening their design skills and improving visual consistency.

Web & Design Team Member - Chemical Engineering Association, IIT Kharagpur **Aug 2023 to Apr 2024**
- Designed **digital assets** such as event posters and promotional materials for departmental events engaging **500+** students.
- Implemented key website updates that improved usability, **content accessibility**, and the department's digital presentation.

Additional Achievements

- **NSS Volunteer:** Conducted digital literacy & online banking workshops for **200+** rural people, promoting financial literacy.
- **Athletics:** Won **Gold** in **100m** sprint & **Bronze** in **4x100m** relay at **U-18** CBSE Clusters, representing the school.
- **Creative Arts:** Won **Silver** (2024) and **Bronze** (2023) in inter-hall Rangoli competition, showing creative achievement.